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Incorporating Non-NIR Absorbing Agent into Packed Powder Samples in Diffuse Reflectance NIR Measurement to Improve Representation of Sample Composition and Accuracy of Concentration Determination

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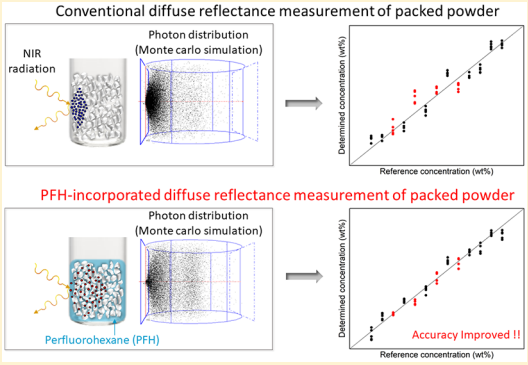
ABSTRACT: A strategy to improve accuracy of near-infrared (NIR) quantitative analysis of powder samples by alleviating the subsampling problem is demonstrated. The approach is to increase spectroscopic sampling volume for more accurate representation of its composition by filling the void space in a packed sample with liquid to reduce differences in refractive index between particle and unoccupied space. By this way, NIR radiation can propagate deeper into the packed powder due to decreased degree of reflection at the interfaces, leading to greater sampling volume. Perfluorohexane (PFH) was chosen as the space-filling liquid since it does not readily absorb NIR radiation and does not disintegrate sample particles due to its strong hydrophobicity. For evaluation, binary mixtures composed of ambroxol and lactose at different concentrations were prepared, and their diffuse reflectance spectra were collected in two ways: conventional direct powder measurement and PFH-incorporated powder measurement. The accuracy of ambroxol concentration determination improved in the PFH-incorporated measurement due to acquisition of more composition-representative spectra by the greater sampling volume. Also, the sample morphology did not change in PFH medium based on XRD analysis. In parallel, Monte Carlo simulation was executed to track NIR photons in the samples and explain the improved accuracy. Improvement in accuracy was also realized when analyzing real pharmaceutical samples containing an active pharmaceutical ingredient in granular form, demonstrating expandability of the proposed scheme.



INTRODUCTION

Diffuse reflectance near-infrared (NIR) spectroscopy has been extensively used for analysis of diverse powder samples[1−5] since its measurement is nondestructive and fast. The adoption of chemometrics[2,4−7] is routine for qualitative or quantitative analysis to extract relevant information since NIR bands of mixtures are typically overlapping and lack of component-specificity due to overtones and combination vibrations. In diffuse reflectance measurement, NIR radiation propagates through a series of particle−air interfaces in a packed sample, and along with absorption, several events such as reflection, refraction, and scattering occur simultaneously at the interfaces.[8,9] These events reduce the chance of forward propagation of photons, and practically irradiated photons actually interact with a small portion of the whole sample packing.[10] Under this situation, an acquired diffuse reflectance spectrum would fall short of representing overall sample composition if some degree of local compositional inhomogeneity existed, which is called the subsampling problem.[2,11,12]

A very viable strategy, unexplored so far, is to increase spectroscopic sampling volume and improve sample representation by replacing void air in the packed sample with liquid for diffuse reflectance measurement. Using this approach, the difference in refractive indices (RIs) between particle and unoccupied volume is decreased, so irradiated photons propagate deeper into the packed sample due to the reduced degree of reflection at interfaces.[13,14] To realize the goal, the space-filling liquid should neither absorb NIR radiation nor alter physical or chemical characteristics of the sample by contact. To meet the requirements, perfluorocarbon (PFC), in which the hydrogens in the hydrocarbon structure are fully substituted with fluorines, has been suggested as a void space-filler because it does not readily absorb NIR radiation and is strongly hydrophobic.[15−17] Since many sample powders such as pharmaceutical ingredients are water-soluble, they would be suspended in a PFC medium without unwanted dissolution, and their physical and chemical characteristics would not be seriously altered.

Initially, perfluorohexane (PFH), perfluorobenzene (PFB), and perfluorononane (PFN) were chosen as potential

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candidates, and their NIR absorption features were investigated to select an optimal liquid. Then, the selected PFC was added to lactose powder, and diffuse reflectance spectra of the lactose/PFC mixture were collected. The acquired NIR spectral features, such as band shape, band intensity, and reproducibility, were examined and compared with those of only lactose powder. Monte Carlo simulation[9,14,18] was employed to examine photon migration in both cases and explain the observed spectral difference. Another issue to consider for PFC-incorporated measurement is potential morphology change of powder in a PFC medium, which would result in variation of sample spectral features and subsequent degradation of accuracy. To address this issue, X-ray diffraction (XRD) diffractograms of pure and PFC-incorporated lactose samples were measured and compared.

Next, binary mixtures composed of ambroxol and lactose with different concentrations were prepared, and NIR spectra of the samples were collected in two ways: conventional direct powder measurement and PFC-incorporated powder measurement. Then, using the collected spectra, partial least-squares (PLS) was employed to determine the concentration of ambroxol in both cases, and subsequent accuracies were compared. To explain the difference in accuracy, a non-homogeneous binary sample system was arbitrarily generated for Monte Carlo simulation, and fluctuation in representation of ambroxol concentration under variations of sample inhomogeneity was evaluated. Finally, real pharmaceutical samples containing an active pharmaceutical ingredient (API) in granular form were measured using the two different experimental schemes, and accuracies were also compared.

■ EXPERIMENTAL SECTION

Sample Preparation and Acquisition of NIR Spectra. Lactose and ambroxol hydrochloride were purchased from Duksan Pharmaceutical Company (Korea) and Tokyo Chemical Industry Company (Japan), respectively. PFH, PFB, and PFN were obtained from Sigma-Aldrich. Thirteen binary mixture (lactose and ambroxol) samples were prepared in this study. For mixing, lactose and ambroxol powders (a total of 6.0 g) were transferred into a 20 mL glass vial, and the vial containing the sample was vigorously shaken by hand over 5 min. The ambroxol concentration ranged from 9.0 to 15.0 wt % with an increment of 0.5 wt %. All 24 Anaprox samples composed of naproxen sodium (API), microcrystalline cellulose (MCC), lactose hydrate, and magnesium stearate were provided by Chong Kun Dang Pharmaceutical Co. (Cheonan, Korea). Naproxen sodium and the other three constituents were in granular and powder forms, respectively. The concentration variations of naproxen sodium, MCC, and lactose hydrate in the samples were 55.5∼82.0, 11.0∼30.7, and 5.0∼11.8 wt %, respectively, and the concentration of magnesium stearate was unvaried at 2.0 wt %.

Diffuse reflectance NIR spectra of ambroxol/lactose mixtures and Anaprox samples were collected using an FT-NIR spectrometer (ABB) equipped with a quartz-halogen light source and an indium arsenide (InAs) detector. A sample-containing 20 mL vial was continuosly rotated during spectral acquisition. The angles of illumination and detection were 45° with respect to the sample-contained vial. Teflon powder (35 μm) was used as a background, and five replicate spectra were collected after randomly repacking the sample each time. For PFH-incorporated measurements, 4.0 and 3.4 mL of PFH were added, respectively, to 6.0 g of ambroxol/lactose mixture and 4.9 g of Anaprox sample in 20 mL vials. The addition of PFH made the sample slurry-like. The PFH-added sample was shaken by hand again over 1 min and tapped several times before measurement. NIR spectra of the PFH-incorporated samples were measured using the same experimental configuration. All collected NIR spectra of the samples were pretreated with multiplicative scatter correction (MSC) before further analysis. MSC and PLS were conducted using MATLAB version 2017a (The Math Works Inc., Natick, MA,).

XRD diffractograms of lactose, ambroxol, and naproxen sodium were obtained using an X-ray diffractometer (Rigaku Model SmartLab) equipped with Cu Kα X-ray radiation operating at 45 kV and 200 mA. The incidence angle varied from 5 to 50° with an increment of 0.01°, and the scan rate was 1° min^{-1}. XRD diffractograms of the same sample mixed with PFH were also acquired.

Monte Carlo Simulation. Monte Carlo simulation was executed using fully licensed Monte Carlo Version 2016 to investigate photon propagation in the ambroxol/lactose sample. The basic assumptions for the simulation were that individual photons independently propagated through a medium in a random walk-like fashion in three-dimensional space, and that incident radiation had a uniform intensity over the illumination area. The detailed parameters used for simulation, such as refractive index, scattering coefficient (μ_s), and absorption coefficient (μ_a), are summarized in Table 1. The optical coefficients of lactose powder were obtained by a linear fit based on previous data acquired by a radially diffused reflectance method.[9] Also, the same parameters were used for ambroxol, since it had similar optical properties to lactose. The parameters for PFH were obtained from a previous publication.[19] The number of illuminated NIR photons was 100 000, and the photons were evenly distributed over the illumination area in the simulation. The diameters of the illumination area and detection window were set to 10 and 30 mm, respectively, which were close to those of the actual experimental configuration. After illumination of photons, the reflected photons, equivalent to photons not absorbed by a sample, were counted at the detection window. Absorbance was calculated as (#incident photons − #reflected photons)/#incident photons, where # indicates number.

Table 1. Parameters of Each Sample Used for Monte Carlo Simulation in This Study

	average particle size (μm)	density (g/cm^3)	refractive index	absorption coefficient, μ_a (cm^{-1})	scattering coefficient, μ_s (cm^{-1})
lactose	24.1	1.525	1.652	35.6	202.6
ambroxol	42.2	1.586	1.622	34.7	209.3
PFH		1.669	1.252	0	140

■ RESULTS AND DISCUSSION

NIR Spectral Features of Lactose Powder and PFH-Incorporated Lactose Powder. Figure 1 shows NIR transmission spectra of PFH, PFB, and PFN in the 9000−4000 cm^{-1} range collected with a path length of 2 mm. The spectra are arbitrarily offset for clear comparison. In the PFB spectrum (black), numerous absorption peaks appear in the 6000−4000 cm^{-1} range; however, their absorbances are not high (maximum absorbance of 0.198 for the strongest peak at

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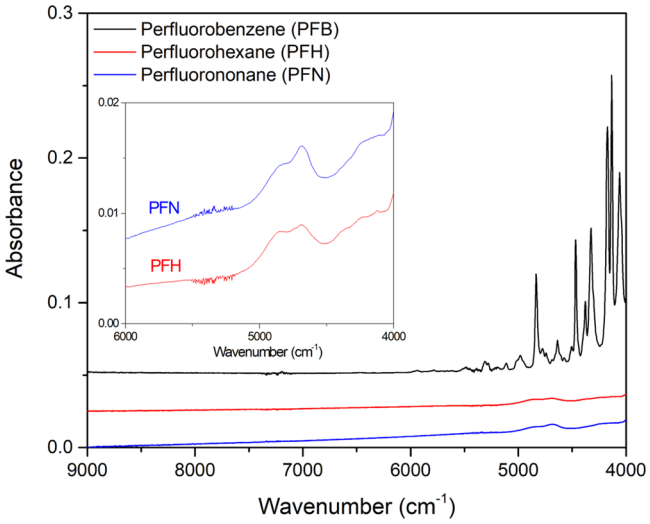

Figure 1. NIR transmission spectra of perfluorobenzene (PFB), perfluorohexane (PFH), and perfluorononane (PFN) in the 9000–4000 cm^{-1} range collected with a path length of 2 mm. NIR peaks of PFH and PFN in the 6000–4000 cm^{-1} range are also highlighted in the inset.

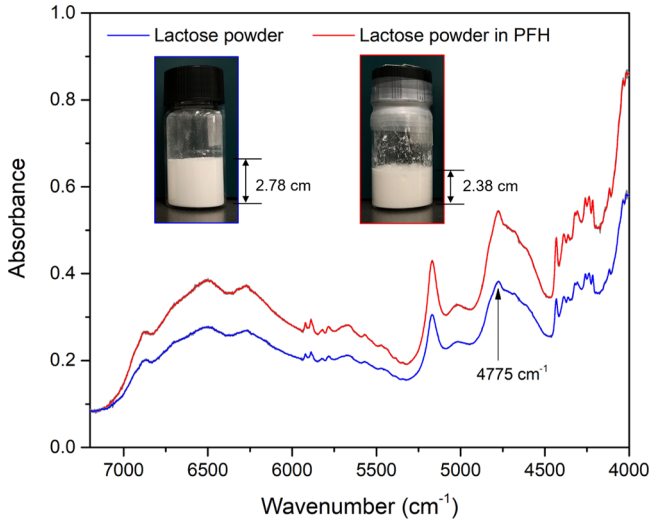

Figure 2. Diffuse reflectance NIR spectra of lactose powder (blue) and lactose/PFH mixture (red). The samples of lactose powder and lactose/PFH mixture in glass vials contain equivalent masses of lactose and are shown with their corresponding packing height.

4133 cm^{-1}) even with use of a thick path length (2 mm). In contrast, no significant absorption peaks are observed in the spectrum of PFH (red) or PFN (blue) over the whole spectral range, although minute absorption peaks are present in the 5000–4000 cm^{-1} range (refer to the inset). The absolute absorbances of these peaks for PFH and PFN are 0.009 and 0.015, respectively, so their NIR absorption will not seriously interfere with diffuse reflectance spectral features of a powder sample.

Although both PFH and PFN were usable from a spectral standpoint, PFN was rejected since its density (1.799 g/cm^3) was considerably higher than that of lactose (1.525 g/cm^3), a dominant component in ambroxol/lactose binary mixture. When PFN and ambroxol/lactose sample (ambroxol concentration: 12.5 wt %) were mixed, a two-phase separation easily occurred due to the large density difference. When the same sample was mixed with PFH (density: 1.669 g/cm^3), no noticeable phase separation was observed. So, PFH was selected as the space-filler for subsequent diffuse reflectance measurements.

Figure 2 shows pictures of lactose powder and lactose/PFH mixture in a glass vial. The packing volume decreases notably from 13.4 to 11.5 cm^3 (14.2% reduction) with addition of PFH. The packing volume decreases due to the more facile rearrangement of individual particles in the slurry phase. The figure also presents MSC-treated diffuse reflectance NIR spectra of lactose powder and lactose/PFH mixture. Five replicate spectra collected by randomly repacking the sample at each spectral acquisition are presented. The overall shapes of the spectra are nearly identical; however, the magnitude of NIR absorption is greater for the lactose/PFH mixture compared to lactose powder. The greater absorption was attributed to the longer penetration depth of NIR radiation caused by the reduced RI difference between lactose (RI: 1.652) and void space saturated with PFH (RI: 1.252). The reduced RI difference resulted in a decrease in the degree of reflection at interfaces and increased penetration depth of NIR radiation into the sample. Next, the reproducibility of each measurement was assessed by calculating relative standard deviation (RSD) of absorbance of the strongest lactose peak at 4775 cm^{-1}. The RSD was 0.07 and 0.10% in direct lactose and PFH-incorporated lactose measurements, respectively. The reproducibility values of the two measurements were good and showed no substantial difference.

Another important issue to address for PFH-incorporated measurement is potential change in morphology of powder in the PFH medium, which can induce variation of sample spectral features and eventual degradation of accuracy. To assess possible sample morphological change, XRD analysis was performed for lactose powder and lactose/PFH mixture, and the acquired diffractograms are shown in Figure 3(a). In the two cases, the strong peak appears at 20.81°, and the XRD patterns are nearly identical. The observation confirms no morphological change of lactose powder in PFH. Since PFH is strongly hydrophobic, its interaction with lactose particles is minimal, and no morphology change of lactose in PFH is thereby consequential.

Monte Carlo Simulation on Propagation of Photons Inside Sample. For Monte Carlo simulation, the total volume was set to 13.4 and 11.5 mL, respectively for lactose powder and PFH-incorporated lactose powder (refer to Figure 2), while the numbers of lactose particles in the two samples were the same. The photon wavelength was 2095 nm, equivalent to 4775 cm^{-1} and corresponding to the most intense lactose peak. Figure 4(a) and (b) show, respectively, the side-view distribution of reflected photons in lactose powder and PFH-incorporated lactose powder when photons are irradiated at 45° with respect to the left side of the packed sample. Figure 4(c) presents the number of photons distributed along the depth inside the sample. The photon distributions differ substantially between cases. In lactose powder, the photons largely populate near the left wall, with the maximal population at a depth of 1.8 mm, and they penetrate to approximately 6.0 mm in depth. Meanwhile, photon penetration in the lactose-PFH mixture reaches 13 mm. The reduced RI difference between the particle and its surroundings is responsible for deeper photon propagation and greater sample interaction volume leading to increased magnitude of absorption peaks, as experimentally demonstrated in Figure 2.

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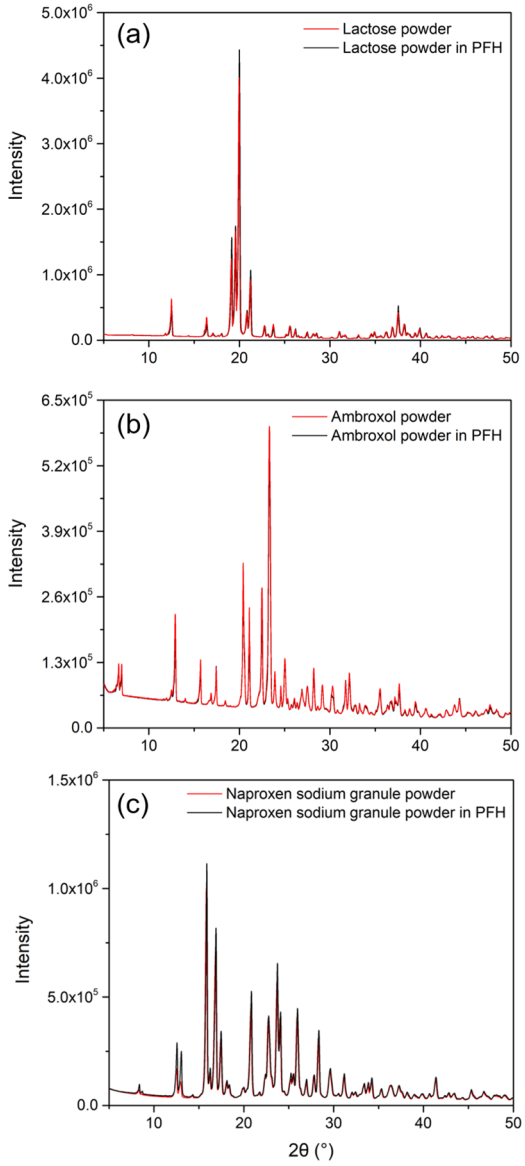

Figure 3. XRD diffractograms of (a) lactose powder and lactose/PFH mixture, (b) ambroxol powder and ambroxol/PFH mixture, and (c) granular naproxen sodium and granular naproxen sodium/PFH mixture.

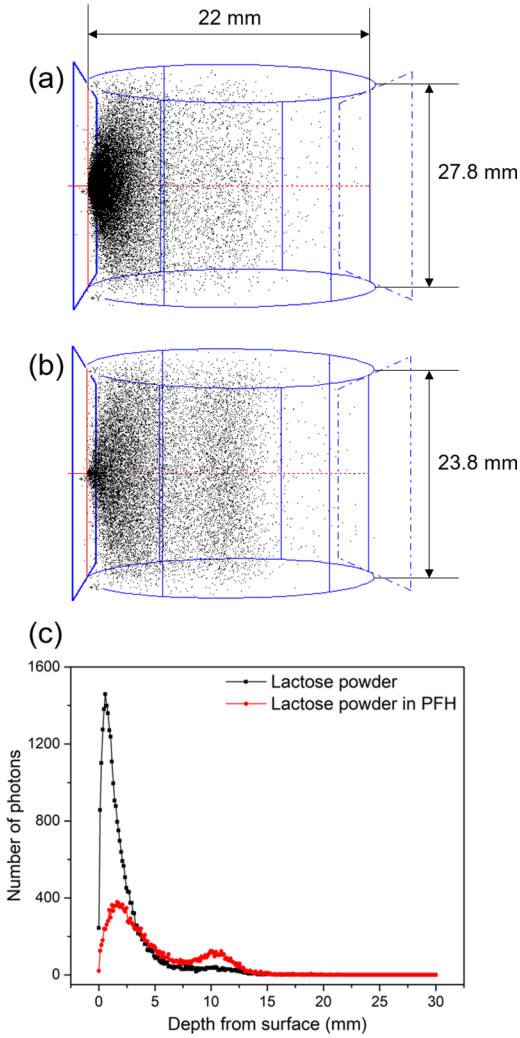

Figure 4. Side-view distribution of reflected photons (black dots) within the lactose powder (a) and PFH-incorporated lactose powder (b) when photons are irradiated at a 45° incident angle with respect to the left side of the packed sample. The numbers of photons distributed along the depth within the sample for the two cases (c) are also shown.

Determination of Ambroxol Concentration in Ambroxol/Lactose Mixture. To evaluate whether the PFH-filling strategy was potential to improve accuracy of NIR quantitative analysis, determination of ambroxol concentration (from 9.0 to 15.0 wt % at increments of 0.5 wt %) in ambroxol/lactose mixtures was attempted. Before evaluation, no morphological change of ambroxol in PFH was confirmed (refer to the corresponding XRD data in Figure 3(b)). Figure 5 shows NIR spectra of 12.5 wt % ambroxol/lactose mixture and the same sample incorporated with PFH. The NIR spectrum of pure ambroxol is also shown to identify its features in the mixture spectra. The ambroxol peak at 5012 cm^{-1} is apparent on the spectral features of lactose in both spectra, and the overall absorption is greater for the PFH-incorporated sample. The five replicate spectra were reproducible in both cases. The average RSDs of the ambroxol peak at 5012 cm^{-1} calculated from the spectra collected from all 13 mixture samples were nearly the same; 0.16 and 0.14% in the direct powder and PFH-incorporated measurements, respectively.

PLS was employed to determine the ambroxol concentrations in the samples using the 7200−4000 cm^{-1} range. Of a total of 13 samples, nine randomly chosen samples (ambroxol concentration: 9.0, 9.5, 10.5, 11.5, 12.5, 13.5, 14.0, 14.5, and 15.0 wt %) were assigned to a calibration set to build a PLS model, and the four remaining samples (ambroxol concentration: 10.0, 11.0, 12.0, and 13.0 wt %) were used as a validation set. Using the calibration sample set, a PLS model was constructed with a leave-one-out cross-validation, and the ambroxol concentrations of validation samples were determined using the developed model. In the case of direct powder measurements, the resulting standard error of cross validation (SECV) was 0.20 wt %, and standard error of prediction (SEP) was 0.60 wt %. Meanwhile, the accuracy improved considerably in the PFH-incorporated measurement (SECV: 0.23 wt %, SEP: 0.26 wt %). In both cases, three principal components were employed in the PLS models. Figures 6(a) and (b) show the concentration correlation plots (reference

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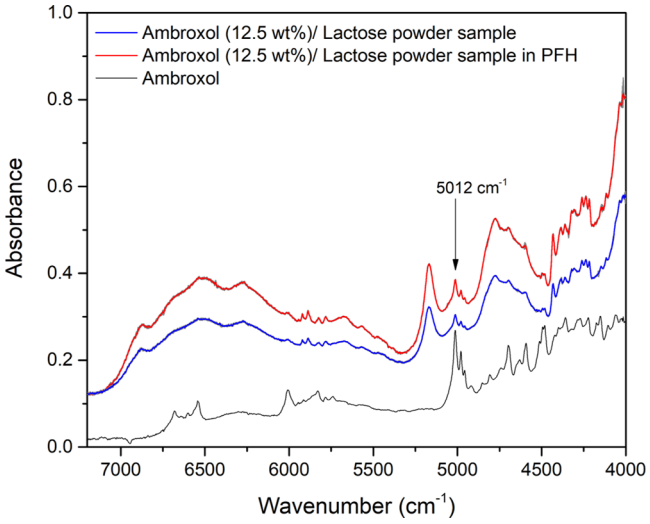

Figure 5. NIR spectra of 12.5 wt % ambroxol/lactose mixture and the same sample mixed with PFH. NIR spectrum of pure ambroxol is also shown to identify its features in the mixture spectra.

concentration vs determined concentration) in both measurements. The data points are much closer to the diagonal line (ideal line) in the PFH-incorporated measurement.

The origin of improved accuracy in the PFH-incorporated measurement merits discussion. A possible factor influencing on the accuracy is spectroscopic sampling volume. Although ambroxol and lactose powders were thoroughly mixed for sample preparation, local mixing inhomogeneity exists in the samples. Under this situation, sampling of whole packed samples in spectral acquisition is ideal to correctly represent the overall composition of mixed components; nonetheless, it is practically unachievable, and most spectra are acquired by partial sampling. The larger spectroscopic sampling volume provided by the PFH-incorporated measurement should facilitate more accurate representation of whole sample composition, thereby yielding more accurate determination of component concentrations.

For deeper investigation, Monte Carlo simulation was performed for an arbitrary sample in which ambroxol was localized at five different locations in ambroxol/lactose mixture, as shown in Figure 7(a). The top view of the binary mixture sample is shown, and five small circles indicate the locations of ambroxol surrounded by lactose. In a three-dimensional view, each circle corresponds to a cylinder occupying 5.17% of the total sample volume. Figures 7(b) and (c) show the photons that interacted with ambroxol inside the #1 cylinder (located at the center of packed sample) in conventional powder and PFH-incorporated measurements, respectively. As expected, the photon population in the #1

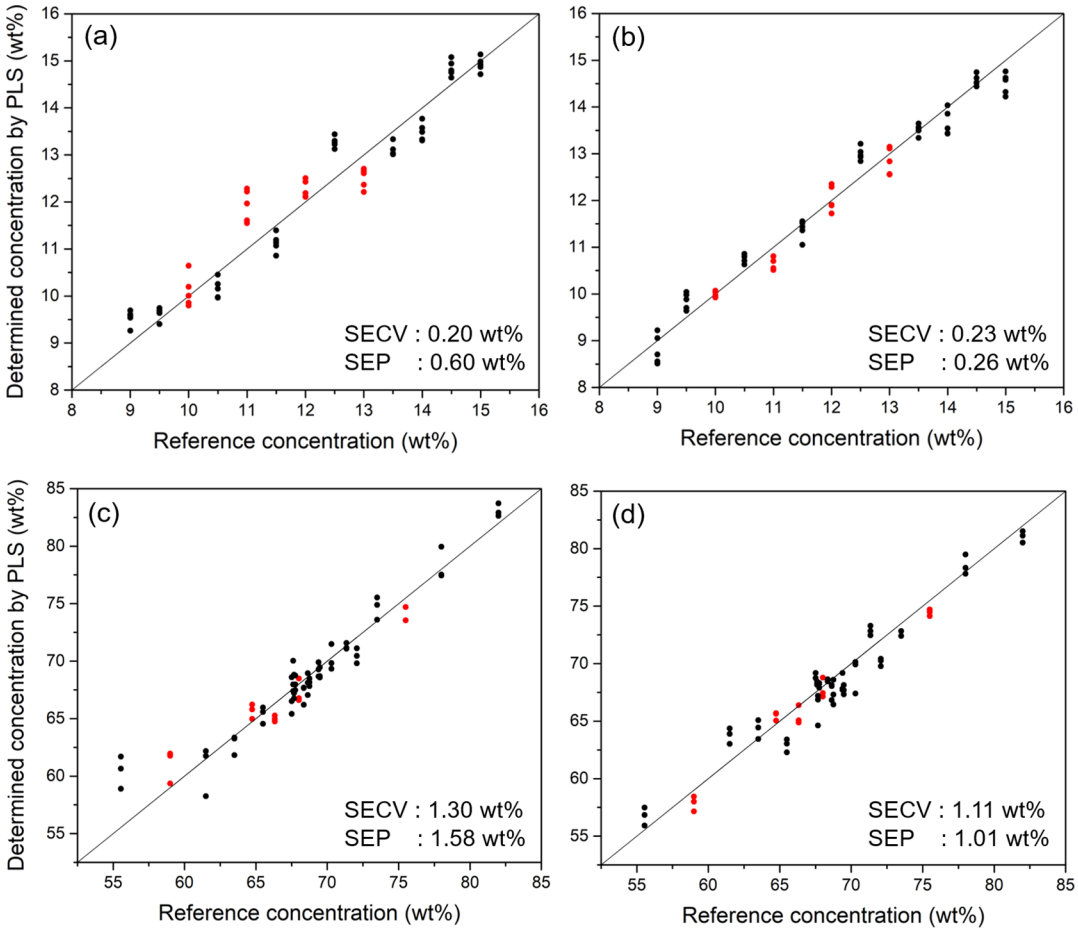

Figure 6. Concentration correlation plots (reference concentration vs determined concentration) resulting from direct ambroxol/lactose powder (a) and PFH-incorporated ambroxol/lactose powder (b) measurements, and direct Anaprox sample (c) and PFH-incorporated Anaprox sample (d) measurements. The black and red circles indicate the calibration and validation samples, respectively.

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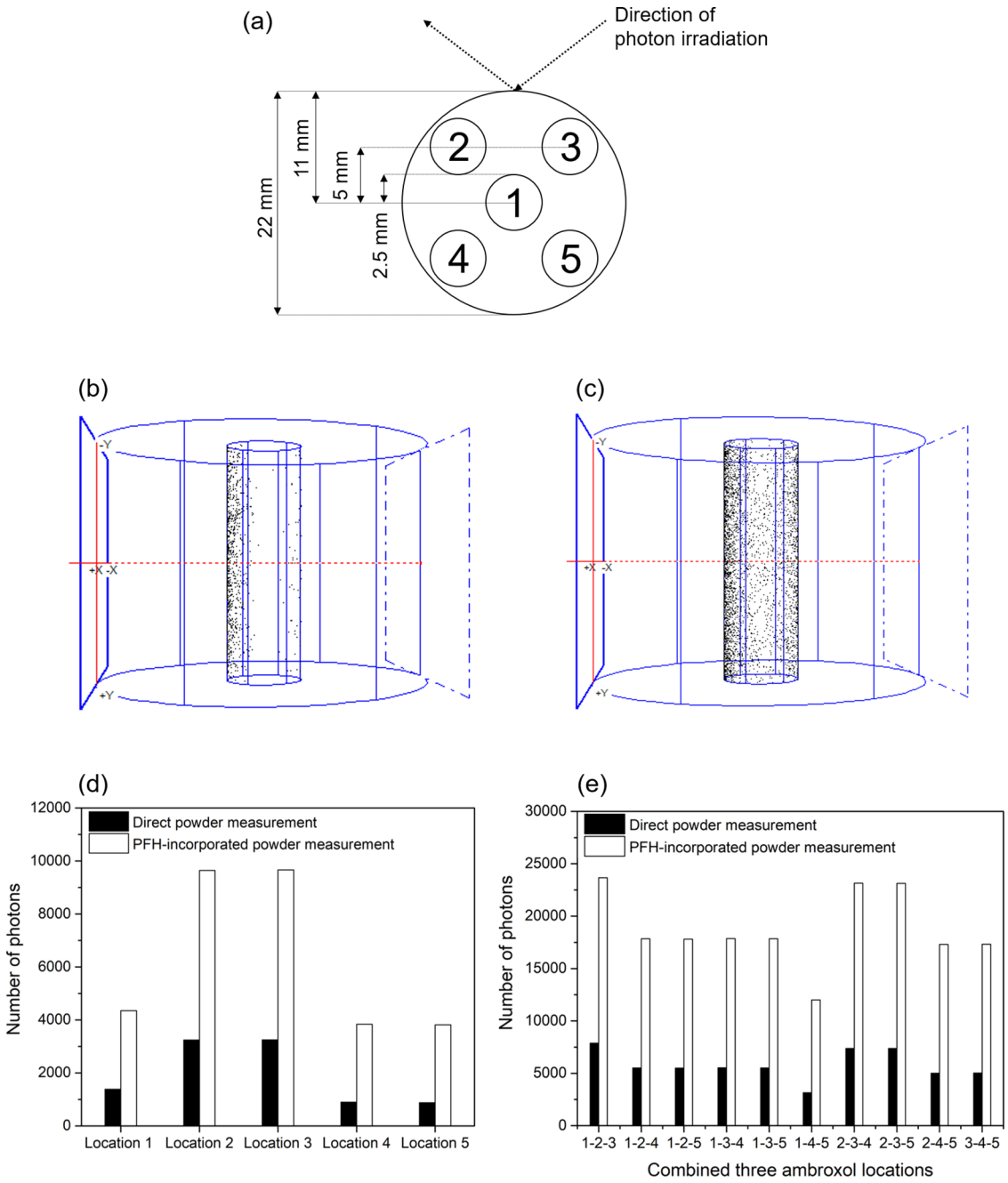

Figure 7. Top view of generated inhomogeneous ambroxol/lactose mixture for Monte Carlo simulation, in which the five circles correspond to the locations of ambroxol (a) and the distribution of photons interacted with ambroxol inside the #1 cylinder in the conventional (b) and PFH-incorporated (c) measurements. The numbers of interacted photons inside each ambroxol cylinder (d) and inside the combined three ambroxol locations (a total of 10 combinations, (e)) in both measurements are shown.

cylinder is larger in the PFH-incorporated measurement due to deeper penetration of NIR photons. Figure 7(d) shows the number of interacted photons in each ambroxol cylinder in both measurements. In each case, photon populations are higher in the #2 and #3 cylinders due to the close proximity of the photon irradiation source. The photon populations at any given cylinder are always larger in the PFH-incorporated measurement.

To emulate the fluctuation of concentration representation when a sample is inhomogeneous, three of the five cylinders were randomly selected (a total of 10 combinations), and the total number of photons from the combined three cylinders was compared across measurements, as shown in Figure 7 (e). The number of interacted photons was largest and smallest in combinations of 1−2−3 and 1−4−5, respectively. The combination of 1−4−5 corresponds to the ambroxol locations farthest from the point of NIR irradiation. The number of interacted photons varies depending on the locations of the three selected sample cylinders.

The calculated RSDs of the numbers of interacted photons from the 10 combinations are 23.1 and 18.1% in the conventional and PFH-incorporated measurements, respectively. Since sampling volume is larger in PFH-incorporated measurements, fluctuation in number of ambroxol-interacted

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photons is relatively smaller under simulated inhomogeneous ambroxol distribution. Based on the simulation results, the large sampling volume helps to reduce fluctuation in representation of component concentration when a sample mixture is not homogeneous. In reality, a perfectly homogeneous powder mixture sample is virtually unobtainable. In this sense, the PFH-incorporated measurement scheme is advantageous to collect diffuse reflectance NIR spectra with more accurate representation of overall ambroxol concentration in the sample, ultimately improving the accuracy of concentration determination.

Expansion for Analysis of Real Pharmaceutical Samples. The PFH-incorporated scheme was further tested for analysis of real pharmaceutical process samples containing some components of granular form,[20−22] such as Anaprox samples containing granular naproxen sodium as an API. As is known, NIR spectral features of a solid sample are different depending on whether it exists either in powder or granular form.[23−25] The most concerning issue is potential disintegration of granular morphology of a sample in PFH, which causes a large decrease in the accuracy of concentration determination. XRD diffractograms of naproxen sodium granules (API in Anaprox sample) and the same sample mixed with PFH were therefore examined (refer to Figure 3(c)). The XRD patterns for the two samples are nearly identical, confirming no disintegration of granular structure of naproxen sodium in PFH.

Figure 8(a) shows NIR spectra of four individual components in Anaprox powder samples. The spectrum of naproxen sodium mixed with PFH is also presented. The 5961 and 5143 cm^{-1} bands of naproxen sodium are from the second overtone of aryl C−H and O−H hydrogen bonding as a result of granulation,[26] respectively. As anticipated from XRD analysis, mixing of naproxen sodium with PFH does not change its spectral features. Overall NIR absorption increases due to deeper penetration of photons as a result of PFH inclusion. Figure 8(b) shows average NIR spectra of all Anaprox samples and the same samples mixed with PFH. The shaded regions overlapped with each average spectrum indicate the standard deviations of absorbance at each wavenumber, equivalent to the magnitude of sample-to-sample spectral variation. The naproxen sodium peak at 5961 cm^{-1} is apparent in both spectra, and overall absorption is again greater for the PFH-incorporated sample.

The naproxen sodium concentrations in the samples were determined by PLS using the 7200−4000 cm^{-1} range. Of a total of 24 samples, 19 were selected as a calibration set, and the five remaining samples were used as a validation set. In both cases, three principal components were employed in the PLS models. Figures 6(c) and (d) show the concentration correlation plots resulting from direct powder and PFH-incorporated powder measurements, respectively. The predicted concentrations are closer to the diagonal line and less scattered in PFH-incorporated measurement. SECV and SEP in the direct powder measurement were 1.30 and 1.58 wt %, respectively, while the accuracy improved in the PFH-incorporated measurements (SECV: 1.11 wt %, SEP: 1.01 wt %).

■ CONCLUSIONS

The demonstrated PFH-incorporated NIR measurement scheme was effective to increase the spectroscopic sampling volume of powder mixture sample and was able to acquire a more component concentration-representative diffuse reflectance spectrum, which in turn enabled more accurate concentration determination. The accuracy improvement was also realized in analysis of real pharmaceutical samples containing API in granular form. The scheme was executable by simply mixing a sample with PFH. Therefore, we are confident that PFH-incorporated NIR diffuse reflectance-based compositional analysis can be expanded not only to pharmaceutical samples, but also many different powder samples in diverse fields. Thus, the proposed scheme could be one of new techniques which have seldom proposed over the last several years in the field of NIR spectroscopy.

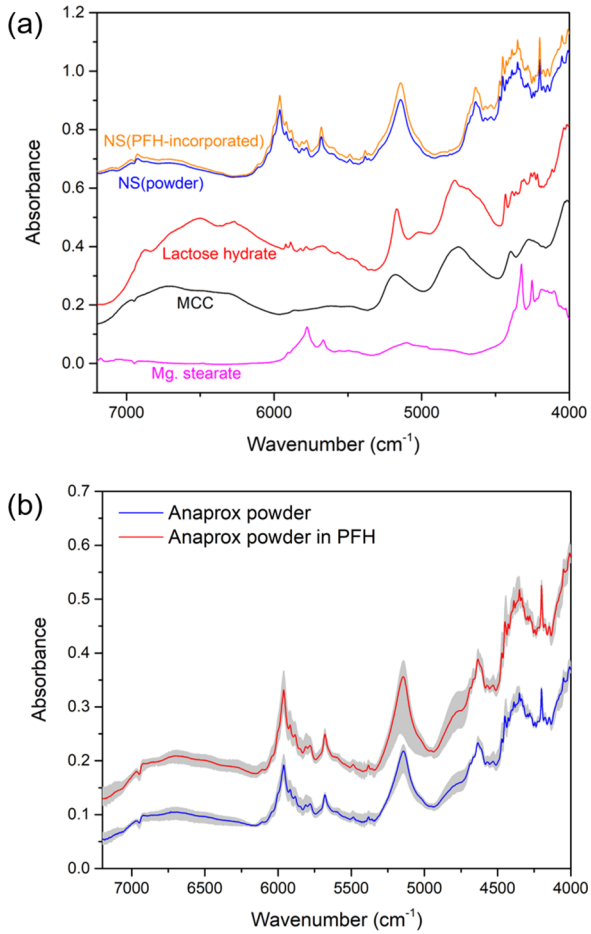

Figure 8. NIR spectra of each component in Anaprox powder samples and PFH-incorporated naproxen sodium (a). Average spectrum of all the Anaprox powder samples and the same samples mixed with PFH are shown (b). The shaded regions indicate the standard deviations of absorbance at each wavenumber.

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Notes

The authors declare no competing financial interest.

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